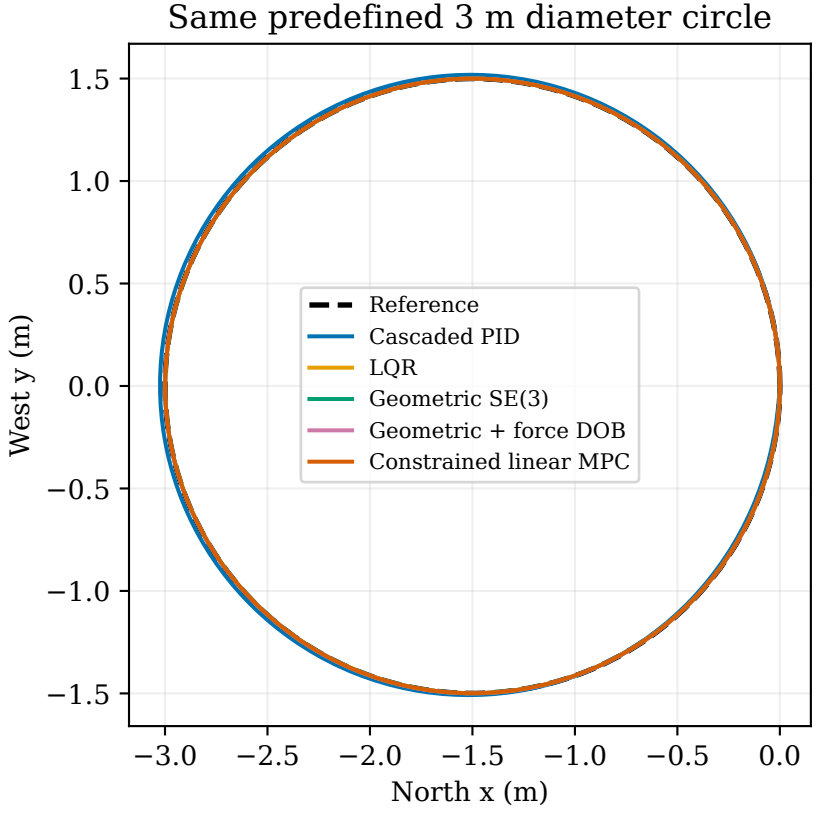
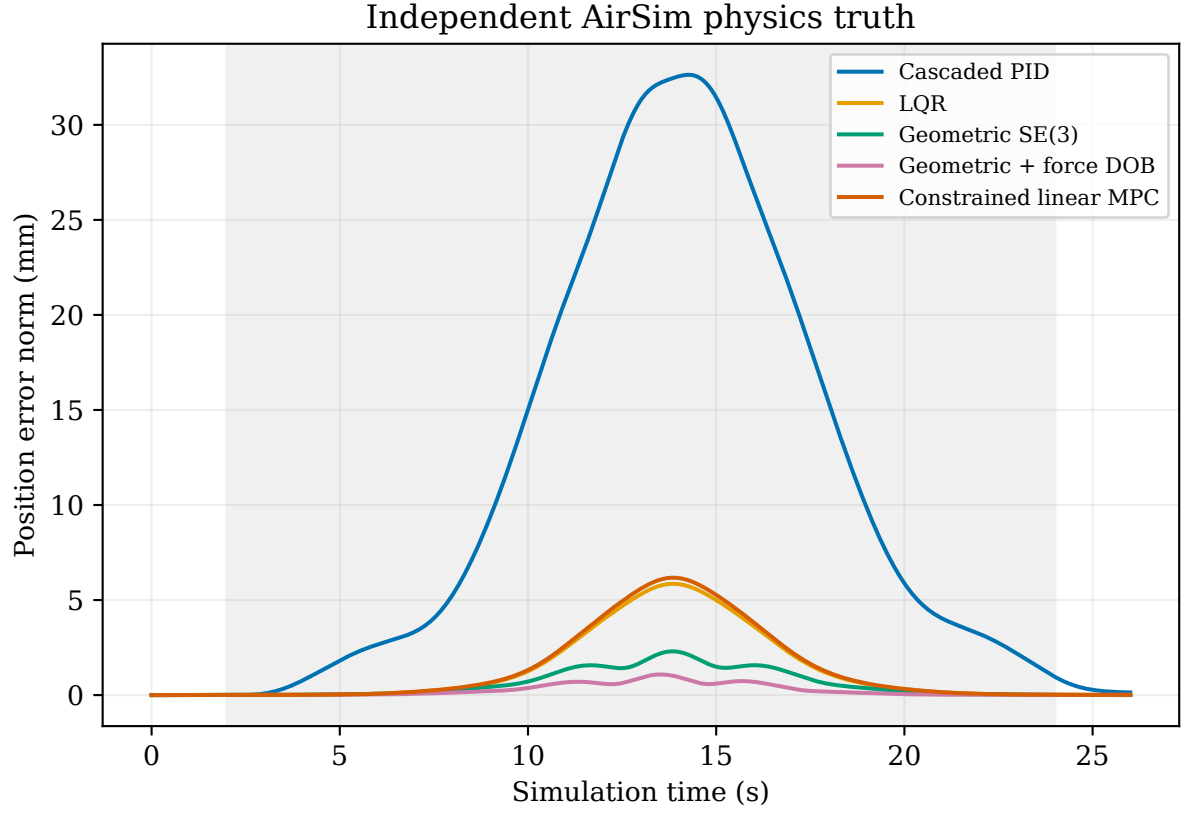

Same predefined 3 m diameter circle
Reference
Cascaded PID
LQR
Geometric SE(3)
Geometric + force DOB
Constrained linear MPC
West y (m)
North x (m)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0


Independent AirSim physics truth
Cascaded PID
LQR
Geometric SE(3)
Geometric + force DOB
Constrained linear MPC
Position error norm (mm)
Simulation time (s)
30
25
20
15
10
5
0
0
5
10
15
20
25